Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/23/2010
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Objective Monitoring of Crew Neurobehavioral Functions (105-day Russian Chamber Study)		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/01/2009	**End Date:**	07/31/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2010 (from 01/31/2011) per NSBRI (09/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Metaxas, Dimitris (Rutgers University) Basner, Mathias (University of Pennsylvania)		
Grant/Contract No.:	NCC 9-58-NBPF00004		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project deployed 2 objective neurobehavioral monitoring techniques (psychomotor vigilance This project deployed two objective neurobehavioral monitoring techniques (psychomotor vigilance test (PVT); facial video for optical computer recognition (OCR) and 2 mood scales (Beck Depression Inventory (BDI); Positive and Negative Affect Schedule (PANAS) in the 105-day Russian chamber study at the Institute for Biomedical Problems (IBMP) in Moscow, Russia. The IBMP chamber consisted of interconnected sealed modules to simulate a space craft (analog) environment that included confinement, isolation, time delays, pressurization, dependence on and autonomy from mission control, and related behavioral factors. The 105-day isolation experiment in the chamber was the second of IBMP's three chamber studies (14-day study, 105-day study, and 520-day study), which collectively are referred to by IBMP as the Mars-500 project. IBMP indicated that the 105-day chamber study focused on obtaining and analyzing scientific-technical information for optimal organization of preparation and efficient conduct of the main (third) experiment (i.e., 520-day study) simulating all the stages of a manned flight to Mars. Consequently, the 105-day chamber experiment was a pilot study for the 520-day experiment. Lessons learned in the 105-day study were to be used to improve experimental techniques and the completeness and validity of the data acquired in the 520-day study. This project on Objective Monitoring of Crew Neurobehavioral Functions in RCS-105 was one of 72 separate protocols conducted in the IBMP's 105-day chamber study. The overall goals of this project were to (1) determine the feasibility of acquiring neurobehavioral data (facial video, cognitive performance, psychological assessments) on the 6-person crew during the 105-day simulated mission, using specially programmed computers (mission controllers were added to the data acquisition requirements after the study commenced); (2) determine the feasibility of monitoring mood and fatigue states in crew members throughout the 105-day mission; and (3) Determine the reliability of 3-minute PVT performance throughout the 105 days. Acquisition of neurobehavioral data on the 6 crewmembers during the 105-day mission in the IBMP Mars-500 chamber was successfully completed using specially programmed notebook computers. During the simulated mission, crewmembers completed a total of 85 (94%) scheduled Beck Depression Inventories (BDI-II); 78 (87%) scheduled Positive and Negative Affect Schedules (PANAS); 1,104 (88%) Psychomotor Vigilance Tests (PVT) and visual analog scale (VAS) ratings; and 19 (63%) WinSCAT tests. Facial videos of crewmembers to assess feasibility of optical computer recognition (OCR) of emotional expressions and fatigue during the PVT tests yielded only 125 (11%) videos technically adequate for OCR evaluation, due to interference from facility lighting (i.e., face shadowed or too dark). Data from 19 mission controllers showed lower adherence on all measures than did the crewmembers. Controllers completed 63% of the PANAS and BDI-II assessments, and only 50% of PVT. Across time in mission, crewmembers' positive mood states tended to decline modestly, but negative moods states and depression were not evident and did not increase. Similarly, VAS ratings of stress and unhappiness did not change across the 105-day mission. In contrast, VAS mental fatigue declined and alertness improved significantly across time in mission. PVT performance on the standard 10-min. PVT task showed increasing errors across time in mission, but the optimized 3-min. PVT did not, nor did measures from the WinSCAT performance battery. Mission controllers showed no statistically significant changes in either positive or negative mood states or performance across time in mission except for total errors on the 10-min. PVT, which appeared to be due to increasing lax effort on the task. Night shift work by crewmembers increased subjective (VAS) ratings of sleepiness, tiredness and physical exhaustion, and increased performance errors, on the PVT, especially the 3-min. optimized PVT. The 105-day project results identified the following changes to procedures that would be needed to ensure the success of such data in longer-duration analog studies (e.g., IBMP's 520-day Mars mission simulation). 1. Adherence to the neurobehavioral procedures in the special computer used need to be adjusted to reduce the evaluation time burden on crews. 2. Evaluation of data quality and quantity must occur regularly throughout the mission to reduce data loss by identifying and correcting problems. 3. Acquisition of facial video for OCR from the built-in computer camera is feasible in the chamber if environmental lighting is standardized, and if special portable lighting of the face in the plane of the computer screen is used during the testing, when video is acquired. 4. Acquisition of subjective reports should use brief rating scales, assess interpersonal conflicts, and by include a measure of social desirability bias at pre-mission. 5. Actigraphy should be used 24/7 during the mission to provide an objective measure of crew behavioral rest-activity cycles for interpretation of performance and subjective data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The 3-minute PVT Self Test used in the 105-day chamber study is being developed to help people quickly and objectively detect the extent to which fatigue is affecting their alertness and reaction times. As such, the technology has high potential for usefulness in a range of safety-sensitive environments on Earth. Potentially any occupation in which alertness and fatigue management are essential to prevent errors on critical tasks will benefit from adaptations of the PVT Self Test technology (e.g., airport security screeners, physicians on night shifts and prolonged call, etc.). The optical computer recognition (OCR) technology being developed for objective monitoring and management of stress, negative emotional states and fatigue in space flight has applications for many Earth-based safety-sensitive occupations, such as transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel.
	A total of 1,104 10-minute Psychomotor Vigilance Tasks (PVT) and 1,104 3-minute PVTs (PVT Self test) were acquired from the 6 crewmembers over the course of the 105-day chamber confinement. All 2,208 PVT test bouts completed have been extracted and reduced to key outcome variables. The 18 mission controllers completed 978 10-minute and 978 3-minute PVT trials, which have been extracted and reduced to key outcome variables. A total of 1,129 facial videos (acquired from the six crewmembers during the 3-minute PVTs) were evaluated for feasibility of analysis by optical computer recognition (OCR). A total of 1,010 facial videos (acquired from the 18 mission controllers during the 3-minute PVTs) also were evaluated for feasibility of analysis by optical computer recognition (OCR). The crew and mission controllers completed the Beck Depression Inventory (BDI), the Positive and Negative Affect Schedule (PANAS), and the Spaceflight Cognitive Assessment Tool for Windows (WINSCAT). A total of 85 scheduled BDI questionnaires and 26 unscheduled BDI questionnaires were acquired from the crew (totaling 111 BDIs), all of which have been extracted and reduced. A total of 135 BDI questionnaires were acquired from mission controllers and are being extracted. A total of 48 scheduled PANAS questionnaires and 60 additional PANAS questionnaires were acquired from the crew (totaling 108 PANAS), all of which have been extracted and reduced. A total of 156 PANAS questionnaires were acquired from mission controllers and are being extracted. A total of 19 scheduled WinSCAT bouts were acquired from the crewmembers; all of which have been extracted and reduced. Mission controllers were not requested to perform the WinSCAT due to operational constraints.

Task Progress:	SUPPLEMENTAL REPORTING FOR FINAL REPORT (August 2010): Acquisition of neurobehavioral data on the 6 crewmembers during the 105-day mission in the IBMP Mars-500 chamber was successfully completed using specially programmed notebook computers. During the simulated mission, crewmembers completed a total of 85 (94%) of scheduled Beck Depression Inventories (BDI-II); 78 (87%) of scheduled Positive and Negative Affect Schedules (PANAS); 1,104 (88%) Psychomotor Vigilance Tests (PVT) and visual analog scale (VAS) ratings; and 19 (63%) of the WinSCAT tests. Facial videos of crewmembers to assess feasibility of optical computer recognition (OCR) of emotional expressions and fatigue during the PVT tests yielded only 125 (11%) videos technically adequate for OCR evaluation, due to interference from facility lighting (i.e., face shadowed or too dark). Data from 19 mission controllers showed lower adherence on all measures than did the crewmembers. Controllers completed 63% of the PANAS and BDI-II assessments, and only 50% of PVT tests. Across time in mission, crewmembers' positive mood states tended to decline modestly, but negative moods states and depression were not evident and did not increase. Similarly, VAS ratings of stress and unhappiness did not change across the 105-day mission. In contrast, VAS mental fatigue declined and alertness improved significantly across time in mission. PVT performance on the standard 10-min. PVT task showed increasing errors across time in mission, but the optimized 3-min. PVT did not, nor did measures from the WinSCAT performance battery. Mission controllers showed no reliable changes in positive or negative mood states or performance across time in mission. Night shift work by crewmembers increased subjective (VAS) ratings of sleepiness, tiredness and physical exhaustion, and increased performance errors on the PVT, especially the 3-min. PVT. The 105-day project results identified the following changes to procedures that would be needed to ensure the success of such data in longer-duration analog studies (e.g., IBMP's 520-day Mars mission simulation). 1. Adherence to the neurobehavioral procedures in the special computer used need to be adjusted to reduce the evaluation time burden on crews. 2. Evaluation of data quality and quantity must occur regularly throughout the mission to reduce data loss by identifying and correcting problems. 3. Acquisition of facial video for OCR from the built-in computer camera is feasible in the chamber if environmental lighting is standardized, and if special portable lighting of the face in the plane of the computer screen is used during the testing, when video is acquired. 4. Acquisition of subjective reports should use brief rating scales, assess interpersonal conflicts, and by include a measure of social desirability bias at pre-mission. 5. Actigraphy should be used 24/7 during the mission to provide an objective measure of crew behavioral rest-activity cycles for interpretation of performance and subjective data.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Goel N, Rao H, Durmer JS, Dinges DF. "Neurocognitive consequences of sleep deprivation." Semin Neurol. 2009 Sep;29(4):320-39. PMID: 19742409 , Sep-2009
Articles in Peer-reviewed Journals	Lim J, Wu WC, Wang J, Detre JA, Dinges DF, Rao H. "Imaging brain fatigue from sustained mental workload: an ASL perfusion study of the time-on-task effect." Neuroimage. 2010 Feb 15;49(4):3426-35. PMID: 19925871 , Feb-2010
Articles in Peer-reviewed Journals	Lim J, Dinges DF. "A meta-analysis of the impact of short-term sleep deprivation on cognitive variables." Psychol Bull. 2010 May;136(3):375-89. PMID: 20438143 , May-2010
Articles in Peer-reviewed Journals	Mollicone DJ, Van Dongen HP, Rogers NL, Banks S, Dinges DF. "Time of day effects on neurobehavioral performance during chronic sleep restriction." Aviat Space Environ Med. 2010 Aug;81(8):735-44. PMID: 20681233 , Aug-2010
Awards	Dinges DF. "Elected to full membership in the International Academy of Astronautics, July 2010." Jul-2010
Awards	Dinges DF. "Raymond F. Longacre Award for Outstanding Accomplishment in the Psychological and Psychiatric Aspects of Aerospace Medicine, May 2009." May-2009